THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 5: **Parallel and Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Lines $l \parallel m$. If $(3x + 10)^\circ$ and $(2x + 20)^\circ$ are co-interior, the larger angle is: 1

(a) 120° **(b)** 100° **(c)** 90° **(d)** 110°

2. In a pattern, $\angle A$ is supplementary to $\angle B$ and $\angle B$ is complementary to $\angle C = 20^\circ$. $\angle A =$ 1

(a) 100° **(b)** 120° **(c)** 130° **(d)** 110°

3. The angle between the bisectors of the two angles of a linear pair is: 1

(a) 180° **(b)** 90° **(c)** 45° **(d)** 60°

4. A transversal cuts parallel lines; the interior angle on one side is $(3x)^\circ$ and the alternate angle is $(x + 30)^\circ$. The angle (in degrees) is: 1

(a) 55° **(b)** 35° **(c)** 45° **(d)** 65°

5. Two angles are supplementary; the difference is 20°. The larger is: 1

(a) 110° **(b)** 90° **(c)** 120° **(d)** 100°

6. Two angles are complementary; the ratio is $1 : 2$. The larger is: 1

(a) 60° **(b)** 70° **(c)** 80° **(d)** 50°

7. If a transversal is perpendicular to one of two parallel lines, it is also: 1

(a) equal to it **(b)** parallel to it

(c) bisecting it **(d)** perpendicular to the other

8. Lines $l \parallel m$. If $(3x + 5)^\circ$ and $(2x + 25)^\circ$ are co-interior, the larger angle is: 1

(a) 85° **(b)** 95° **(c)** 115° **(d)** 105°

9. In a pattern, $\angle A$ is supplementary to $\angle B$ and $\angle B$ is complementary to $\angle C = 35^\circ$. $\angle A =$ 1

(a) 115° **(b)** 145° **(c)** 125° **(d)** 135°

10. If $l \parallel m$, the bisectors of a pair of alternate interior angles are: 1

(a) perpendicular **(b)** intersecting at 45°

(c) parallel **(d)** equal in length

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Two lines intersect forming angles a and b (vertically opposite) with c adjacent to both. Show that $a = b$ using linear pairs, and verify with $c = 40^\circ$. 2

12. Lines $l \parallel m$; $\angle 1 = (5x - 10)^\circ$ and $\angle 2 = (4x + 15)^\circ$ are corresponding angles. Find x and $\angle 1$. 2

13. Find the angle whose supplement is 3 times its complement. 2

14. Two lines intersect forming angles a and b (vertically opposite) with c adjacent to both. Show that $a = b$ using linear pairs, and verify with $c = 75^\circ$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A transversal makes angles $(2x + 10)^\circ$ and $(x + 40)^\circ$ with parallel lines as alternate exterior angles (equal for parallel lines). Find x, then the co-interior-supplementary partner of the angle. 3

16. Angles on one side of a straight line are x°, $(x + 10)^\circ$, $(2x - 2)^\circ$ and 20°. Find x and each angle. 3

17. Show that if two parallel lines are cut by a transversal, the bisectors of a pair of alternate interior angles are parallel. Verify with the angle 40° (each half 20°). 3

OR

A transversal makes angles $(2x + 15)^\circ$ and $(x + 55)^\circ$ with parallel lines as alternate exterior angles (equal for parallel lines). Find x, then the co-interior-supplementary partner of the angle. 3

18. Angles on one side of a straight line are x°, $(x + 12)^\circ$, $(2x - 4)^\circ$ and 24°. Find x and each angle. 3

OR

Show that if two parallel lines are cut by a transversal, the bisectors of a pair of alternate interior angles are parallel. Verify with the angle 50° (each half 25°). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Lines $l \parallel m$ and transversal t. $\angle 1 = (4x + 10)^\circ$ and $\angle 2 = (5x - 1)^\circ$ are co-interior angles. A second transversal gives $\angle 3$ corresponding to $\angle 1$. Find x, all three angles and state reasons. 5

OR

Lines $l \parallel m$ are cut by a transversal; two co-interior angles are in the ratio $1 : 2$. Find both angles, the corresponding angle of the larger one at the other line, and the vertically opposite angle of the smaller one. 5

20. Lines $l \parallel m$ and transversal t. $\angle 1 = (4x + 20)^\circ$ and $\angle 2 = (5x - 2)^\circ$ are co-interior angles. A second transversal gives $\angle 3$ corresponding to $\angle 1$. Find x, all three angles and state reasons. 5

OR

Lines $l \parallel m$ are cut by a transversal; two co-interior angles are in the ratio $2 : 3$. Find both angles, the corresponding angle of the larger one at the other line, and the vertically opposite angle of the smaller one. 5

END OF QUESTION PAPER — **Total: 40 Marks**